ENTERPRISE AI ARCHITECTURE • SPECIFICATION & PEDAGOGY

The 3-Tier Enterprise Agent Stack

Side-by-Side Study Guide: Production SKILL.md Specifications vs. 4th-Grader Mental Models

Learn how to architect, structure, and document agentic skills with rigorous engineering contracts and intuitive metaphors.

TIER 0: CLOUD PRIMITIVES

Skills #1 – #7

Physical AWS hardware, DynamoDB CAS locks, S3 WORM vaults, Firecracker microVMs, and EventBridge buses.

TIER 1: AGENT OS KERNEL

Skills #8 – #14

Individual robot brain: Prefrontal propose-decide gates, DAG runbooks, mutex leases, context pagers, and fact detectives.

TIER 2: MULTI-AGENT SWARMS

Skills #15 – #20

Social clubhouse governance: Watts-Strogatz small worlds, liaison ambassadors, bounded hops, shared whiteboards & quorums.

THE SKILL AUTHORING ROSETTA STONE: HOW PLAYGROUND ANALOGIES MAP TO CODE

4TH-GRADER CARD COMPONENT	INTUITIVE PLAYGROUND METAPHOR	PRODUCTION SKILL.MD EQUIVALENT	RIGOROUS ENGINEERING CONTRACT
Hero Persona & Title	Meet Buzzy the Robot / Talking Stick	`YAML Frontmatter (name, version)`	Unique identifier, tier level, privilege ring.
One-Sentence Pitch	Super-fast imagination machine	`YAML 'description' field`	Semantic summary indexed by router models.
Playground Recess Story	Buzzy writes on paper with hands locked	`1. Purpose & Architectural Boundary`	Strict operational boundary; separation of concerns.
Toy Box Flowchart	Idea Box -> Paper Blueprint -> Safe World	`2. Interface Contract (JSON Schema)`	Typed JSON schema validating inputs & returns.
The Golden Rule	'Wait for the stick before your turn!'	`3. Core System Invariants`	Non-negotiable invariants (CAS locks, zero IAM write).
Page 22 Wire Connection	Blue wire plugs down into Safe #3	`4. Execution Flow & Syscalls`	Deterministic state machine and downstream calls.
Superpower	Crushes hallucinations with 3 clues	`5. Failure Modes & Self-Healing`	SLAs, timeout caps, fallback rollback handlers.

PLAYGROUND MENTAL MODEL (FOR 4TH GRADERS)

Meet Buzzy the Idea Robot!

WHAT IS THIS SKILL IN ONE SENTENCE?

A super-fast imagination machine that comes up with cool ideas, but isn't allowed to touch anything in real life!

The Playground Story (How It Works):

Imagine you have a robot friend named Buzzy who is amazing at writing mystery stories and dreaming up giant LEGO castles. Whenever you ask him a question, Buzzy writes down a blueprint on paper. But Buzzy has NO hands! He cannot touch scissors, paint, or real coins. He is plugged into the Master Blueprint on Page 21: he hands his blueprint paper up to Officer Cedar and Kernel Propose-Decide. If they don't approve it, the idea stays on paper forever!



MAP CONNECTION TO PAGE 22: Plugs into Kernel Propose-Decide (Skill #8)

Buzzy's blue wire feeds ideas straight up into the Brain Judge (#8). The Judge reviews the blueprint before any hands are unlocked!

Superpower: Super-fast brainstorming without any risk of breaking real toys!

The 4th Grader Golden Rule:

"The Idea Maker proposes, but never pushes the real button!"

PRODUCTION SKILL.MD SPECIFICATION (ENGINEERING CONTRACT)

```
---
name: aws-bedrock
description: Foundation model proposal engine generating struct
tier: Level 0 (Cloud Primitives)
ring: Ring 3 (Untrusted Proposal Space)
permissions: [bedrock:InvokeModel]
tools: [synthesize_proposal, validate_schema]
---
```

1. Purpose & Architectural Boundary

Inference engine in Ring 3; proposes state mutations with 0 IAM write access.

2. Core System Invariants (Non-Negotiable)

- Zero Direct Execution: Never mutates databases or invokes live APIs direc
- Deterministic Schema: Output must strictly validate against JSON Schema.
- Quarantined Payloads: All outputs marked RING3_PROPOSAL until kernel appr

3. Execution Protocol & Syscalls

1. Receives natural language objective & response schema from Kernel (#8).
2. Invokes bedrock:InvokeModel with temperature <= 0.2.
3. Validates output JSON against typed contract; quarantines payload.
4. Hands proposal to Prefrontal Kernel Judge (#8) for verification.

4. Failure Modes & Self-Healing

- Schema Parse Error: Returns REJECTED_SCHEMA for structured retry (max 2).
- Rate Limit (429): Exponential backoff with full jitter (capped at 3 retri

PLAYGROUND MENTAL MODEL (FOR 4TH GRADERS)

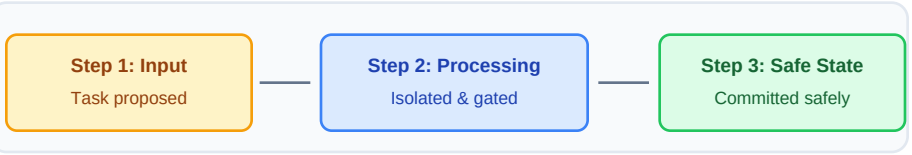
Meet Officer Cedar the Hall Monitor!

WHAT IS THIS SKILL IN ONE SENTENCE?

The strict playground security guard who checks your hall pass before you can enter the room!

The Playground Story (How It Works):

Remember that strict hall monitor who stands outside the principal's office? Even if an AI robot runs up shouting 'I have an emergency!', Officer Cedar stops it, puts his hand out, and asks: 'Do you have a signed Cedar hall pass?' On Page 21's Blueprint, Officer Cedar stands right between untrusted ideas and real cloud computers. If the pass says NO, the door stays locked tight. Zero exceptions, not even for smart robots!



MAP CONNECTION TO PAGE 22: Plugs into Kernel Propose-Decide (Skill #8)

The blue wire connects Officer Cedar to the Brain Judge (#8). Every time the Judge evaluates an action, Cedar inspects the hall pass!

Superpower: Blocks unauthorized moves before anything bad can happen!

The 4th Grader Golden Rule:

"No hall pass? No entry. Zero rule breaking allowed!"

PRODUCTION SKILL.MD SPECIFICATION (ENGINEERING CONTRACT)

```
---
name: aws-agentcore
description: Cedar policy guardrail gate evaluating authorizati
tier: Level 0 (Cloud Primitives)
ring: Ring 0 (Deterministic Gatekeeper)
permissions: [agentcore:EvaluatePolicy, cedar:Authorize]
tools: [evaluate_guardrail, inspect_syscall]
---
```

1. Purpose & Architectural Boundary

Authoritative gate evaluating Cedar AST policies against untrusted proposals.

2. Core System Invariants (Non-Negotiable)

- Default Deny: Unmatched principal/action/resource triples are blocked.
- Sub-10ms Latency: In-memory AST evaluation before any network I/O.
- Immutable Rules: Cedar policy files sealed in versioned repositories.

3. Execution Protocol & Syscalls

1. Intercepts proposed syscall from Kernel Propose-Decide (#8).
2. Evaluates Cedar policy AST: Permit/Forbid on (Principal, Action, Resourc
3. If FORBID: Emits immediate Ring 0 Security Violation alarm.
4. If PERMIT: Generates signed ephemeral authorization token.

4. Failure Modes & Self-Healing

- Unauthorized Syscall: Rejection logged to CloudWatch audit trail.
- Syntax Error in Policy: Fails closed (DENY) to prevent security holes.

PLAYGROUND MENTAL MODEL (FOR 4TH GRADERS)

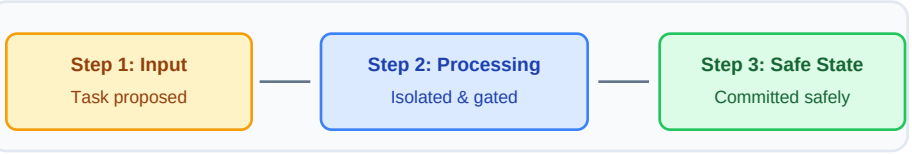
Meet The Golden Fortress Safe!

WHAT IS THIS SKILL IN ONE SENTENCE?

A magical metal safe where once you drop a diary page in, nobody on earth can ever erase it!

The Playground Story (How It Works):

Have you ever made a promise and wished you could carve it in solid diamond so no one could ever lie about it later? That is S3 WORM Vault! When an AI finishes a test or scores a soccer goal, it writes down the receipt and drops it through a one-way mail slot into a titanium safe. The padlock is frozen for 7 whole years. On Page 21, look at all the blue wires plugging into this safe: the Context Pager (#12), the Wax Diary (#13), and the 3-Friend Fact Checker (#14) all store their golden receipts here!



MAP CONNECTION TO PAGE 22: Plugs into Context Pager (#12), Episodic Ledger (#13), & Grounding

Three separate blue wires plug into this safe! Memory frames, diary pages, and certified facts are all locked under 7-year WORM compliance.

Superpower: Guarantees 100% honesty because you cannot change history!

The 4th Grader Golden Rule:

"Once it is dropped inside the safe, it stays forever!"

PRODUCTION SKILL.MD SPECIFICATION (ENGINEERING CONTRACT)

```
---
name: aws-s3-vault
description: Immutable WORM receipt vault with Object Lock comp
tier: Level 0 (Cloud Primitives)
ring: Ring 0 (Physical Hardware Boundary)
permissions: [s3:PutObject, s3:GetObjectLockLegalHold]
tools: [store_compliance_receipt, get_immutable_fact]
---
```

1. Purpose & Architectural Boundary

Cryptographic WORM storage ensuring past execution records cannot be altered.

2. Core System Invariants (Non-Negotiable)

- Strict WORM Compliance: S3 Object Lock retains objects for >= 7 years.
- SHA-256 Checksums: Every receipt verified via Content-MD5 / SHA-256.
- No Overwrites: Versioning enabled; s3:DeleteObject strictly forbidden.

3. Execution Protocol & Syscalls

1. Ingests execution receipts from Context Pager (#12) & Episodic Ledger (#
2. Calculates cryptographic SHA-256 digest of payload.
3. Stores object under s3://agent-vault/{agent_id}/{timestamp}.json with Ob
4. Returns immutable ARN receipt used by Grounding Certifier (#14).

4. Failure Modes & Self-Healing

- Checksum Mismatch: Aborts transaction; logs data corruption event.
- Lock Expiry Attempt: S3 compliance mode denies administrative overrides.

PLAYGROUND MENTAL MODEL (FOR 4TH GRADERS)

Meet Locker-Bot with 3 Secret Drawers!

WHAT IS THIS SKILL IN ONE SENTENCE?

A super-smart locker that holds keys, detective maps, and memory drawers so toys don't get mixed up!

The Playground Story (How It Works):

Imagine your school locker has 3 magic drawers. Drawer #1 has the Talking Stick Key (DynamoDB CAS Mutex): whoever grabs it gets the soccer ball, and nobody else can touch it! Drawer #2 has a magnifying glass for finding matching cards. Drawer #3 has a spiderweb map of friendships. On Page 21's Blueprint, the blue wire connects Drawer #1 straight up to the Talking Stick Keeper (#11) so kids never collide!



MAP CONNECTION TO PAGE 22: Plugs into Kernel Mutex (#11) & Grounding Certifier (#14)

The blue wire provides atomic DynamoDB locks to the Talking Stick (#11), and vector/graph facts to Detective Triangulator (#14).

Superpower: Instantly remembers facts and stops two kids from fighting over t

The 4th Grader Golden Rule:

"One key per kid at a time = zero arguments!"

PRODUCTION SKILL.MD SPECIFICATION (ENGINEERING CONTRACT)

```
---
name: aws-database
description: DynamoDB CAS mutex, OpenSearch vectors & Neptune g
tier: Level 0 (Cloud Primitives)
ring: Ring 0 (State & Storage Engine)
permissions: [dynamodb:PutItem, opensearch:Search, neptune:Quer
tools: [cas_acquire_lock, vector_knn_query, graph_triple_
---
```

1. Purpose & Architectural Boundary

Tri-store architecture: atomic locks, semantic embeddings, and graph relations.

2. Core System Invariants (Non-Negotiable)

- Atomic CAS Concurrency: Mutex writes require conditional version check.
- Vector Similarity Bound: Cosine similarity threshold >= 0.82 for facts.
- Graph Integrity: Neptune entity relations must satisfy valid ontology.

3. Execution Protocol & Syscalls

1. DynamoDB: Provides conditional write primitives for Kernel Mutex (#11).
2. OpenSearch: Returns top-k semantic chunks with distance metrics.
3. Neptune: Evaluates entity-relationship triples for Detective Triangulato
4. Emits verified state snapshots to the central agent kernel.

4. Failure Modes & Self-Healing

- ConditionalCheckFailed: DynamoDB indicates active lock contention.
- Low Vector Score: Returns NO_SEMANTIC_MATCH to prevent hallucinated groun

PLAYGROUND MENTAL MODEL (FOR 4TH GRADERS)

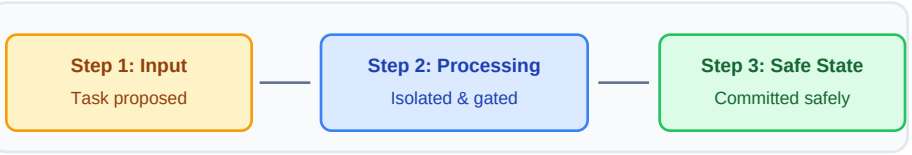
Meet The 10-Second Pop-Up Tent!

WHAT IS THIS SKILL IN ONE SENTENCE?

A temporary play-tent that pops up in 1 second to do your math homework, then disappears completely!

The Playground Story (How It Works):

What if you had a pop-up tent that you threw on the grass whenever you had math homework? You hop inside, use the calculator, solve 50 fractions in 2 seconds, and then *POOF!* The tent folds itself up and vanishes into thin air. No messy crumbs from yesterday can ever spill into today! On Page 21, the Runbook Engine (#9) and Watchdog Sparky (#10) pitch this pop-up tent whenever a calculation needs to be solved safely away from the rest of the school.



MAP CONNECTION TO PAGE 22: Plugs into Runbook Engine (#9) & Watchdog Supervisor (#10)

The blue wire sends tasks into isolated Firecracker microVM tents. Watchdog (#10) keeps a 15-second timer on the tent!

Superpower: Fresh, clean sandbox workspace every single time with zero mess l

The 4th Grader Golden Rule:

"Do the math, pack up the tent, leave no mess!"

PRODUCTION SKILL.MD SPECIFICATION (ENGINEERING CONTRACT)

```yaml
---
name: aws-lambda
description: Isolated Firecracker microVM sandboxes for statele
tier: Level 0 (Cloud Primitives)
ring: Ring 3 (Isolated MicroVM Sandbox)
permissions: [lambda:InvokeFunction]
tools: [invoke_sandboxed_tool, terminate_microvm]
---
```

1. Purpose & Architectural Boundary

Zero-footprint ephemeral compute running calculations in Firecracker microVMs.

2. Core System Invariants (Non-Negotiable)

- Strict Ephemerality: MicroVM disposed after task; 0 disk persistence.
- Hard Wall-Clock Cap: Execution forcibly terminated after 15 seconds.
- Resource Sandboxing: Memory capped at 512MB; isolated VPC subnet.

3. Execution Protocol & Syscalls

1. Runbook Engine (#9) compiles atomic task into Python payload.
2. Invokes Lambda function synchronously inside isolated Firecracker sandbo
3. Watchdog Supervisor (#10) starts external countdown timer.
4. Collects stdout & return value; tears down sandbox immediately.

4. Failure Modes & Self-Healing

- Timeout (15s): Lambda runtime terminates process; supervisor alerts swarm
- Memory Exhaustion (OOM): Returns OUT_OF_MEMORY error to Runbook Engine.

AWS Step Functions: The Board Game Referee

Side-by-Side Study: 4th-Grader Mental Model vs. Production SKILL.md Specification

PLAYGROUND MENTAL MODEL (FOR 4TH GRADERS)

Meet The Hopscotch Referee!

WHAT IS THIS SKILL IN ONE SENTENCE?

A board game referee who makes sure you hop on Step 1, then Step 2, and never skip ahead to Step 5!

The Playground Story (How It Works):

Playing Candy Land or Hopscotch is only fun when everyone follows the steps in order. Step Functions is the master referee. It holds a checklist: 'Step 1: Wash hands. Step 2: Eat lunch. Step 3: Brush teeth.' If you try to skip ahead, the referee blows his whistle: *TWEET!* 'Go back to Step 1!' On Page 21, look at the blue wire from the LEGO Assembly Guide (#9): it relies on Step Functions to enforce the exact build sequence so the robot never puts the spaceship roof on backwards!



MAP CONNECTION TO PAGE 22: Plugs into Kernel Runbook Engine (#9)

The blue wire gives the LEGO Guide (#9) strict state machine checkpoints. If an error occurs, it rolls back to the previous safe square.

Superpower: Guarantees you never skip a step or lose your place in a recipe!

The 4th Grader Golden Rule:

"Step 1 before Step 2, every single time!"

PRODUCTION SKILL.MD SPECIFICATION (ENGINEERING CONTRACT)

```
---
name: aws-stepfunctions
description: Deterministic hierarchical finite state machine or
tier: Level 0 (Cloud Primitives)
ring: Ring 0 (Deterministic State Machine)
permissions: [states:StartExecution, states:DescribeExecution]
tools: [start_state_machine, advance_checkpoint, rollback
---
```

1. Purpose & Architectural Boundary

Strict state transition engine guaranteeing runbooks follow exact recipes.

2. Core System Invariants (Non-Negotiable)

- Forward-Only Progression: Steps cannot skip ahead without prerequisite su
- Automatic Compensation: Step failures trigger registered rollback handler
- Exact Event History: Every transition logged with monotonic sequence ID.

3. Execution Protocol & Syscalls

1. Kernel Runbook Engine (#9) deploys Amazon States Language (ASL) definiti
2. Advances through tasks: Step 1 -> Checkpoint -> Step 2 -> Checkpoint.
3. Verifies each output payload against JSON Schema before next transition.
4. Emits completion event to EventBridge Walkie-Talkie (#7).

4. Failure Modes & Self-Healing

- Step Task Failure: Executes Catch / Retry blocks with exponential backoff
- Irrecoverable Error: Triggers compensatory rollback to previous safe stat

PLAYGROUND MENTAL MODEL (FOR 4TH GRADERS)

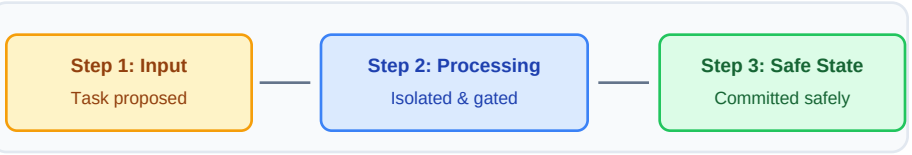
Meet Walkie-Talkie Dispatcher!

WHAT IS THIS SKILL IN ONE SENTENCE?

A high-speed walkie-talkie network that delivers secret messages across the entire school playground!

The Playground Story (How It Works):

Imagine you are on the swings and your friend is way over on the soccer field. How do you tell them recess is ending? You use the playground walkie-talkie! EventBridge listens for news: 'Attention, lunch is ready!' and instantly zips that message to only the kids who need to hear it without screaming on the loudspeaker. On Page 21's Blueprint, Watchdog Sparky (#10) uses this radio wire to sound emergency alarms if a robot gets dizzy!



MAP CONNECTION TO PAGE 22: Plugs into Kernel Supervisor Watchdog (#10)

The blue wire connects the radio bus to Watchdog (#10). If an agent trips a circuit breaker, EventBridge radios all pods instantly!

Superpower: Delivers the right message to the right robot without shouting!

The 4th Grader Golden Rule:

"Whisper to your friends, don't scream on the loudspeaker!"

PRODUCTION SKILL.MD SPECIFICATION (ENGINEERING CONTRACT)

```
---
name: aws-client-api
description: Amazon EventBridge pub/sub message bus and API Gat
tier: Level 0 (Cloud Primitives)
ring: Ring 0 (Perimeter Ingress & Egress)
permissions: [events:PutEvents, apigateway:POST]
tools: [publish_swarm_event, broadcast_emergency_alarm]
---
```

1. Purpose & Architectural Boundary

High-throughput asynchronous event bus routing signals without broadcast shouting.

2. Core System Invariants (Non-Negotiable)

- Targeted Routing: Event patterns match specific pods; 0 global broadcast
- At-Least-Once Delivery: Events buffered in DLQ if consumer is temporarily
- Strict OpenAPI Schema: Payload shape enforced at the API perimeter.

3. Execution Protocol & Syscalls

1. Kernel Supervisor (#10) or Swarm Router publishes event to bus.
2. EventBridge evaluates pattern filter: { detail-type: 'SwarmTaskDispatche
3. Delivers event directly to subscriber pod's dedicated queue.
4. Emits metrics to CloudWatch for monitoring latency and hop counts.

4. Failure Modes & Self-Healing

- Consumer Unreachable: Retries 3 times before routing to Dead Letter Queue
- Malformed Payload: Rejected at ingress with HTTP 400 Bad Request.

PLAYGROUND MENTAL MODEL (FOR 4TH GRADERS)

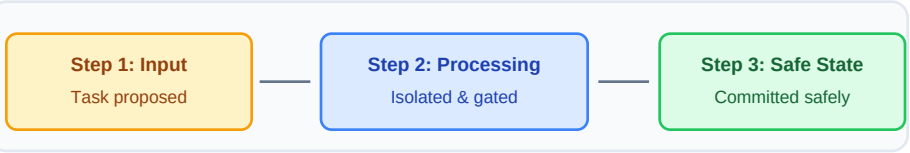
Meet The Prefrontal Brain Judge!

WHAT IS THIS SKILL IN ONE SENTENCE?

Your inner coach that stops you from yelling silly things and makes sure you only do smart moves!

The Playground Story (How It Works):

Have you ever had a silly thought: 'What if I jump off the couch wearing cardboard wings?' That wild idea came from Ring 3. Before your legs jump, your inner Brain Judge (Ring 0) chimed in: 'Wait, that will hurt! Build a pillow fort instead.' Now look at Page 21! Look at the wires connecting to this Judge: Buzzy Bot (#1) feeds ideas up into it, Officer Cedar (#2) verifies the rules, and Ambassador Leo (#16) brings notes from other classrooms to this Judge before taking any action!



MAP CONNECTION TO PAGE 22: Plugs into Bedrock (#1), AgentCore (#2), & Liaison Router (#16)

Two blue wires go down to Bedrock (#1) and Cedar (#2). One purple wire comes down from Ambassador Leo (#16) with pod notes!

Superpower: Filters out silly impulses so the robot never does something dang

The 4th Grader Golden Rule:

"Think before you jump!"

PRODUCTION SKILL.MD SPECIFICATION (ENGINEERING CONTRACT)

```
---
name: kernel-propose-decide
description: Prefrontal executive arbiter enforcing Ring 0 verd
tier: Level 1 (Agent OS Kernel)
ring: Ring 0 (Executive Control)
permissions: [kernel:EvaluateProposal, kernel:DispatchSyscall]
tools: [evaluate_model_proposal, commit_kernel_verdict]
---
```

1. Purpose & Architectural Boundary

Separation of inference (Ring 3) from executive authority (Ring 0): Action = Verdict(Proposal).

2. Core System Invariants (Non-Negotiable)

- Zero Direct Mutation: Ring 3 models hold 0 IAM credentials.
- Mandatory Quorum Gate: Requires active Cedar (#2) & CAS lock (#11).
- Strict Type Safety: Payloads failing JSON Schema trigger immediate halt.

3. Execution Protocol & Syscalls

1. Ingests proposed tool call JSON from Bedrock (#1) or Liaison Router (#16
2. Validates payload against JSON Schema and active goal constraints.
3. Verifies Cedar guardrails (#2) and DynamoDB CAS mutex lease (#11).
4. Dispatches verified syscall to Level 0 AWS hardware; logs verdict to S3.

4. Failure Modes & Self-Healing

- Schema Parse Failure: Returns REJECTED_SCHEMA for retry (max 2).
- Policy Denial: Emits security alert and evicts bad proposal to ledger (#1

PLAYGROUND MENTAL MODEL (FOR 4TH GRADERS)

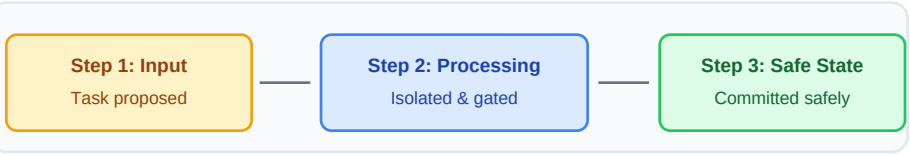
Meet The Master Builder Guide!

WHAT IS THIS SKILL IN ONE SENTENCE?

A colorful step-by-step LEGO instruction booklet that shows you which brick to snap on next!

The Playground Story (How It Works):

When you open a giant Millennium Falcon LEGO box with 2,000 pieces, you don't just dump all the pieces on the rug and guess. You open Book 1, Page 1! You snap the yellow brick to the blue brick. Once that page is done, you turn to Page 2. On Page 21's Blueprint, the Playground Map (#15) feeds its mission goals down into this LEGO booklet, and this booklet uses Step Functions (#6) and the Pop-Up Tent (#5) to build each step safely and in order!



MAP CONNECTION TO PAGE 22: Plugs into Topology (#15), Step Functions (#6), & Lambda (#5)

Connected to Topology (#15) from above, and plugs into Step Functions (#6) & Lambda (#5) below to assemble the LEGO bricks.

Superpower: Turns giant complicated missions into simple, foolproof steps!

The 4th Grader Golden Rule:

"Follow the booklet step-by-step to build a masterpiece!"

PRODUCTION SKILL.MD SPECIFICATION (ENGINEERING CONTRACT)

```
---
name: kernel-runbook-engine
description: Directed Acyclic Graph (DAG) procedural execution
tier: Level 1 (Agent OS Kernel)
ring: Ring 0 (Procedural Orchestrator)
dependencies: [aws-stepfunctions, aws-lambda]
tools: [compile_mission_dag, execute_step_barrier]
---
```

1. Purpose & Architectural Boundary

Decomposes complex objectives into verifiable DAG runbooks with step barriers.

2. Core System Invariants (Non-Negotiable)

- Topological Order: No task executes until all parent nodes succeed.
- Step Barrier Verification: Each phase output verified before next begins.
- Reentrancy: Idempotent step keys allow safe resuming after reboot.

3. Execution Protocol & Syscalls

1. Receives multi-pod mission topology from Swarm Topology Builder (#15).
2. Compiles dependency DAG with clear step checkpoints.
3. Dispatches individual steps to Step Functions (#6) & Lambda (#5).
4. Records milestone completion receipts to Episodic Ledger (#13).

4. Failure Modes & Self-Healing

- DAG Node Failure: Halts branch execution; triggers compensatory rollback.
- Cyclic Dependency: Static DAG validator rejects circular graphs at compil

PLAYGROUND MENTAL MODEL (FOR 4TH GRADERS)

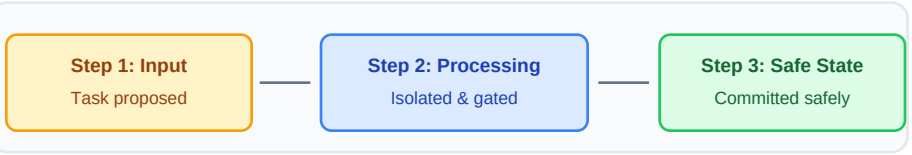
Meet Watchdog Sparky!

WHAT IS THIS SKILL IN ONE SENTENCE?

A friendly watchdog who taps the robot on the shoulder if it gets stuck running in circles!

The Playground Story (How It Works):

Have you ever seen a puppy chase its own tail around in a dizzy circle? If nobody stops him, he falls over! Sometimes, an AI gets stuck saying: 'Uh, what was I doing?' over and over again. Watchdog Sparky watches with an egg timer. If the robot spins 3 times, Sparky barks *WOOF!*, pulls the lever, and says: 'Time out!' On Page 21, the Hot Potato Relay (#17) and Red Card Referee (#20) both plug into Sparky to make sure dizzy robots are stopped!



MAP CONNECTION TO PAGE 22: Plugs into Hop Limiter (#17), Circuit Breaker (#20), & EventBridge (#7

Takes orders from Hop Limiter (#17) & Red Card (#20). Plugs into EventBridge (#7) below to radio the alarm across the school!

Superpower: Stops runaway loops and keeps battery energy from being wasted!

The 4th Grader Golden Rule:

"If you are dizzy, stop and take a breath!"

PRODUCTION SKILL.MD SPECIFICATION (ENGINEERING CONTRACT)

```
---
name: kernel-supervisor-circuit
description: Execution watchdog monitoring loops, timeouts, and
tier: Level 1 (Agent OS Kernel)
ring: Ring 0 (Supervisor Control)
dependencies: [aws-client-api, aws-lambda]
tools: [monitor_heartbeat, trip_circuit_breaker]
---
```

1. Purpose & Architectural Boundary

Watchdog supervisor enforcing timeouts and tripping circuit breakers on infinite loops.

2. Core System Invariants (Non-Negotiable)

- Timeout Ceiling: Any single execution cycle hard-capped at 15 seconds.
- Deadlock Detection: Repeated identical proposals trigger loop interrupt.
- Unconditional Stop: Supervisor signal overrides active agent execution.

3. Execution Protocol & Syscalls

1. Arm countdown egg timer when agent starts task.
2. Ingests tick signals from Hop Limiter (#17) and Circuit Breaker (#20).
3. If loop detected or timeout exceeded: trips breaker and stops agent.
4. Radios emergency alert to EventBridge (#7) and triggers clean reset.

4. Failure Modes & Self-Healing

- Agent Unresponsive: Forcefully terminates Firecracker MicroVM container.
- Cascade Loop: Quarantines agent pod and notifies swarm coordinator.

PLAYGROUND MENTAL MODEL (FOR 4TH GRADERS)

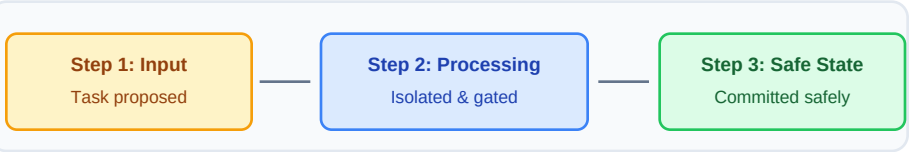
Meet The Talking Stick Keeper!

WHAT IS THIS SKILL IN ONE SENTENCE?

In circle time, only the kid holding the magic stick is allowed to speak, so nobody talks at the same time!

The Playground Story (How It Works):

In Kindergarten circle time, what happens if all 25 kids try to tell their show-and-tell stories at the exact same second? Total noisy chaos! Nobody hears anything! That is why teacher gives out the Talking Stick. Whoever holds the wooden stick speaks for 15 seconds. On Page 21, the Talking Stick plugs directly down into Drawer #1 of the AWS Locker (#4). And from above, both the Giant Chalkboard (#18) and the Red Card (#20) use this stick so agents never scribble on each other's work!



MAP CONNECTION TO PAGE 22: Plugs into AWS Database (#4), Whiteboard (#18), & Circuit Breaker (#

Plugs into Drawer #1 of DynamoDB (#4) below for physical locks. Whiteboard (#18) and Red Card (#20) borrow this stick above!

Superpower: Eliminates noisy collisions and accidental overwrites forever!

The 4th Grader Golden Rule:

"Wait for the stick before you take your turn!"

PRODUCTION SKILL.MD SPECIFICATION (ENGINEERING CONTRACT)

```
---
name: kernel-mutex-lease
description: Distributed CAS mutex coordinator with automatic T
tier: Level 1 (Agent OS Kernel)
ring: Ring 0 (Concurrency Control)
dependencies: [aws-database]
tools: [acquire_lease, heartbeat_lease, release_lease]
---
```

1. Purpose & Architectural Boundary

Distributed lock manager preventing race conditions via atomic compare-and-swap (CAS).

2. Core System Invariants (Non-Negotiable)

- Strict Mutual Exclusion: Exactly 1 agent holds the talking stick per reso
- Deadlock Immunity via TTL: Leases auto-expire in 15s if agent crashes.
- Monotonic Fence Tokens: Incremental integer tokens reject stale out-of-or

3. Execution Protocol & Syscalls

1. Agent requests write access to shared resource (e.g. Whiteboard #18).
2. Executes DynamoDB conditional PutItem with TTL and fence token.
3. While task runs: refreshes TTL heartbeat every 5 seconds.
4. Releases lease upon completion or revokes immediately if benched by #20.

4. Failure Modes & Self-Healing

- Contention (409): Truncated binary exponential backoff with jitter.
- Abandoned Lease: Automatic TTL expiration dissolves lock safely.

PLAYGROUND MENTAL MODEL (FOR 4TH GRADERS)

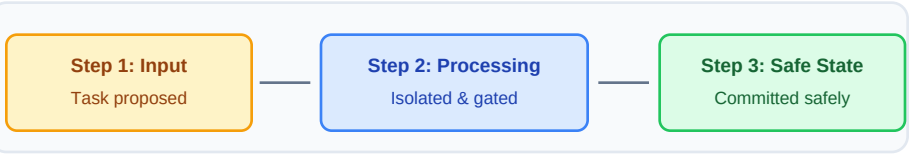
Meet Pocket Backpack Organizer!

WHAT IS THIS SKILL IN ONE SENTENCE?

A magic backpack that keeps only 3 colored pencils in your hands, and packs older markers into the closet!

The Playground Story (How It Works):

If you tried to carry 50 textbooks, 200 crayons, and 5 water bottles in your hands all at once while walking to class, you would drop everything! An AI's working brain can only hold a few thoughts at a time. The Context Pager is a pocket organizer: it keeps the pencil you need right now in your shirt pocket, and neatly packs older homework into the S3 Golden Safe (#3). On Page 21, the blue wire shows this exact pipe: cold thoughts slide into the safe!



MAP CONNECTION TO PAGE 22: Plugs into Amazon S3 Vault (Skill #3)

The blue wire evicts cold reasoning frames down into the 7-year S3 WORM safe (#3), preserving the active 8k token attention window.

Superpower: Keeps your working hands light and sharp with zero memory clutter

The 4th Grader Golden Rule:

"Keep only what you need right now in your hands!"

PRODUCTION SKILL.MD SPECIFICATION (ENGINEERING CONTRACT)

```
---
name: kernel-context-pager
description: Virtual memory manager paging inactive frames to S
tier: Level 1 (Agent OS Kernel)
ring: Ring 0 (Memory Paging System)
dependencies: [aws-s3-vault]
tools: [page_out_cold_frames, page_in_frame, compact_cont
---
```

1. Purpose & Architectural Boundary

Virtual memory paging keeping working attention window sharp (8k tokens) without amnesia.

2. Core System Invariants (Non-Negotiable)

- Bounded Attention Window: Active RAM capped at 8,192 tokens.
- Zero Data Loss: Evicted cold frames stored immutably in S3 (#3).
- Fast Hydration: Paged frames retrieved by key in < 25 milliseconds.

3. Execution Protocol & Syscalls

1. Monitors active token count during reasoning loop.
2. When token threshold >= 75%: evicts oldest reasoning frames to S3 Vault
3. Leaves lightweight pointer stub: [PAGED: frame_id, s3_uri, summary].
4. Pages frame back into memory only when specific reference is requested.

4. Failure Modes & Self-Healing

- Memory Overflow: Emergency compaction strips scratchpad before LLM crash.
- S3 Read Error: Retrieves verified copy from secondary backup replica.

PLAYGROUND MENTAL MODEL (FOR 4TH GRADERS)

Meet The Unbreakable Stamp Diary!

WHAT IS THIS SKILL IN ONE SENTENCE?

A diary where every page has a secret wax stamp matching the page before it, so no pages can be torn out!

The Playground Story (How It Works):

Imagine you have a detective diary. Every day, you stamp Page 1 with red wax. On Page 2, you melt wax from Page 1 into the new seal. This creates an unbreakable chain: Page 1 ➔ Page 2 ➔ Page 3. If a sneaky villain tried to rip out Page 2, the seal on Page 3 cracks! On Page 21, the Giant Chalkboard (#18) sends all completed team findings down into this Wax Diary, and this Diary seals the block into the S3 Time Capsule (#3) below!

Step 1: Input
Task proposed

Step 2: Processing
Isolated & gated

Step 3: Safe State
Committed safely

MAP CONNECTION TO PAGE 22: Plugs into Whiteboard (#18) & Amazon S3 Vault (#3)

Takes verified findings from the Chalkboard (#18) above, chains them with SHA-256 stamps, and seals them into S3 Vault (#3) below.

Superpower: Makes it impossible for anyone to secretly alter past memories!

The 4th Grader Golden Rule:

"A chain of wax stamps keeps the truth safe!"

PRODUCTION SKILL.MD SPECIFICATION (ENGINEERING CONTRACT)

```
---
name: kernel-episodic-ledger
description: Append-only SHA-256 hash-chained immutable audit l
tier: Level 1 (Agent OS Kernel)
ring: Ring 0 (Audit & Ledger Engine)
dependencies: [aws-s3-vault]
tools: [append_block, verify_chain_integrity, export_audi
---
```

1. Purpose & Architectural Boundary

Cryptographic hash-chain linking every action to its perception: Block(n) = SHA256(Block(n-1) + Data).

2. Core System Invariants (Non-Negotiable)

- Append-Only: No block can ever be modified, erased, or reordered.
- Cryptographic Chaining: Every block embeds previous block's SHA-256 hash.
- WORM Durability: Flushed blocks written to S3 Object Lock vault (#3).

3. Execution Protocol & Syscalls

1. Ingests completed action record from Whiteboard (#18) or Runbook (#9).
2. Combines previous block hash + timestamp + action payload.
3. Computes new SHA-256 digest: seals wax stamp on new page.
4. Pushes immutable block to S3 WORM safe for compliance auditing.

4. Failure Modes & Self-Healing

- Tampering Attempt: Any altered past block breaks all downstream hash chec
- Network Partition: Buffers blocks locally in Ring 0 until S3 link resumes

PLAYGROUND MENTAL MODEL (FOR 4TH GRADERS)

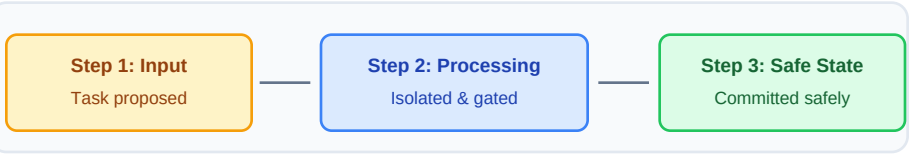
Meet Detective Triangulator!

WHAT IS THIS SKILL IN ONE SENTENCE?

A detective who won't believe a rumor until three different best friends confirm it is 100% true!

The Playground Story (How It Works):

If someone on the playground says: 'Elephants learned how to fly!', you don't instantly believe them! You ask for proof: Friend #1 (S3 Safe #3 photo receipt), Friend #2 (OpenSearch vector cosine match), and Friend #3 (Neptune graph relation in #4). Only when all 3 friends agree does Detective Triangulator say: 'TRUE!' On Page 21, the Supermajority Ballot Box (#19) checks with this Detective before allowing the swarm to cast their votes!



MAP CONNECTION TO PAGE 22: Plugs into Quorum Arbiter (#19), S3 Vault (#3), & Database (#4)

Supplies certified facts up to the Ballot Box (#19), by checking receipts in S3 (#3) and graph triples in AWS Database (#4).

Superpower: Crushes hallucinations by demanding 3 pieces of real proof!

The 4th Grader Golden Rule:

"Never believe a rumor without checking 3 facts!"

PRODUCTION SKILL.MD SPECIFICATION (ENGINEERING CONTRACT)

```
---
name: kernel-grounding-certifier
description: Multi-source fact triangulation requiring 3 indepe
tier: Level 1 (Agent OS Kernel)
ring: Ring 0 (Truth Verification)
dependencies: [aws-s3-vault, aws-database]
tools: [triangulate_fact, verify_quorum_ballot]
---
```

1. Purpose & Architectural Boundary

Crushes hallucinations by demanding 3 independent physical proofs before certifying facts.

2. Core System Invariants (Non-Negotiable)

- Triangulation Rule: Fact certified ONLY when S3 receipt, vector match & g
- Strict Cosine Floor: Vector distance match must exceed 0.82 threshold.
- Deterministic Receipts: Supplies verified cryptographic certificates to #

3. Execution Protocol & Syscalls

1. Ingests candidate fact from model proposal.
2. Checks Proof 1: S3 WORM receipt photo (#3).
3. Checks Proof 2: OpenSearch vector cosine similarity in Database (#4).
4. Checks Proof 3: Neptune entity graph relationship in Database (#4).

4. Failure Modes & Self-Healing

- Unconfirmed Fact: Flagged as UNVERIFIED_RUMOR; discarded from decision pa
- Conflicting Proofs: Rejects proposal and logs conflict to Episodic Ledger

PLAYGROUND MENTAL MODEL (FOR 4TH GRADERS)

Meet The Playground Secret Shortcut Map!

WHAT IS THIS SKILL IN ONE SENTENCE?

A clever clubhouse seating chart where everyone has buddies nearby, plus secret zip-lines across the yard!

The Playground Story (How It Works):

If 12 kids are playing across a huge 3-acre playground, how do you keep them connected without everyone screaming? You split them into 3 cool clubhouses: the Art Team, the Sports Team, and the Science Team. Kids in the same clubhouse whisper easily, and secret zip-lines connect the clubs in just 2 quick hops! On Page 21, look at the purple wire coming out of this map: it plugs directly down into the LEGO Assembly Guide (#9) so the robots know which clubhouse builds first!



MAP CONNECTION TO PAGE 22: Plugs into Kernel Runbook Engine (Skill #9)

The purple wire translates the small-world clubhouse cluster map into step-by-step DAG runbook stages in Skill #9.

Superpower: Connects big teams with short travel distance and zero screaming!

The 4th Grader Golden Rule:

"Sit with your buddies, but keep a zip-line to other clubs!"

PRODUCTION SKILL.MD SPECIFICATION (ENGINEERING CONTRACT)

```
---
name: swarm-topology-builder
description: Watts-Strogatz small-world lattice generator with
tier: Level 2 (Multi-Agent Swarms)
ring: Ring 0 (Swarm Fabric Governance)
dependencies: [kernel-runbook-engine]
tools: [generate_lattice, deploy_liaison_shortcuts, compu
---
```

1. Purpose & Architectural Boundary

Generates small-world graphs (p=0.15): high clustering (C=0.68) with short paths (L=1.8).

2. Core System Invariants (Non-Negotiable)

- Clustered Pods: Specialist agents grouped into tightly-coupled domains (F
- Bounded Diameter: Inter-cluster zip-lines keep network diameter <= 3 hops
- No All-to-All Mesh: Completely eliminates O(N^2) exponential message expl

3. Execution Protocol & Syscalls

1. Organizes 12 agent nodes into 3 specialist pods (Finance, Compliance, Op
2. Establishes dense local connections within each clubhouse.
3. Rewires p=0.15 long-range shortcut bridges between pod liaison leaders.
4. Feeds execution stages down into Kernel Runbook Engine (#9).

4. Failure Modes & Self-Healing

- Network Partition: Adds dynamic backup shortcut bridge if liaison fails.
- Topology Drift: Periodically verifies clustering coefficient and path len

PLAYGROUND MENTAL MODEL (FOR 4TH GRADERS)

Meet Ambassador Leo!

WHAT IS THIS SKILL IN ONE SENTENCE?

The class captain chosen to deliver polite notes between Room 4A and Room 4B without causing a stampede!

The Playground Story (How It Works):

Imagine Mrs. Smith's class wants to challenge Mr. Johnson's class to a kickball game. Should all 30 students run down the hallway at once and barge into Room 4B? No, that would cause a stampede! Instead, the class picks Ambassador Leo. Leo walks politely down the hall, hands the challenge envelope to Room 4B's captain, and brings back the answer. On Page 21, Leo's purple wire connects down to the Prefrontal Brain Judge (#8): every note Leo delivers is vetted before execution!



MAP CONNECTION TO PAGE 22: Plugs into Kernel Propose-Decide (Skill #8)

The purple wire routes inter-pod messages down through the Brain Judge (#8). No message enters a pod without kernel arbitration.

Superpower: Prevents noisy hallway stampedes by sending one smart messenger!

The 4th Grader Golden Rule:

"Send one ambassador instead of the whole mob!"

PRODUCTION SKILL.MD SPECIFICATION (ENGINEERING CONTRACT)

```
---
name: swarm-liaison-router
description: Cross-pod liaison ambassador routing notes without
tier: Level 2 (Multi-Agent Swarms)
ring: Ring 0 (Inter-Pod Routing)
dependencies: [kernel-propose-decide]
tools: [route_cross_pod_message, inspect_pod_envelope]
---
```

1. Purpose & Architectural Boundary

Designates ambassador nodes to mediate inter-pod messaging, preventing broadcast stampedes.

2. Core System Invariants (Non-Negotiable)

- Isolated Pod Boundary: Raw internal pod messages never leak cross-pod.
- Mandatory Kernel Gate: Every inter-pod envelope vetted by Judge (#8).
- Bounded Bandwidth: Rate-limited message exchange via EventBridge bus.

3. Execution Protocol & Syscalls

1. Pod member hands cross-discipline inquiry to designated Liaison Leader.
2. Liaison inspects envelope; packages structured query.
3. Transports message across golden shortcut bridge to peer Liaison.
4. Receiving Liaison passes note through Kernel Propose-Decide (#8) before

4. Failure Modes & Self-Healing

- Unauthorized Sender: Drops unvetted message; logs breach to supervisor.
- Queue Backpressure: Throttles ingress traffic to prevent pod buffer overf

PLAYGROUND MENTAL MODEL (FOR 4TH GRADERS)

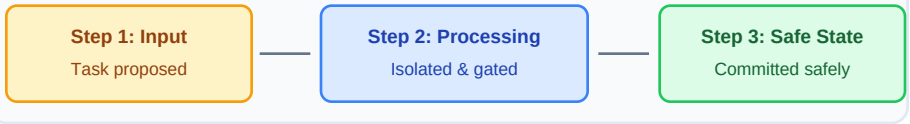
Meet The Hot Potato Referee!

WHAT IS THIS SKILL IN ONE SENTENCE?

A hot-potato referee who pops the balloon if kids keep passing it in circles without doing any work!

The Playground Story (How It Works):

Have you ever played Telephone, where a rumor gets passed from kid to kid until it turns into total gibberish? Or hot potato where kids just pass back and forth forever? The Hop Limiter stamps every message with 3 tickets. Pass 1: Ticket #1 used. Pass 2: Ticket #2 used. Pass 3: Last ticket! If someone tries to pass it a 4th time, the referee shouts: 'Out of tickets! Do the work now!' On Page 21, this ticket puncher connects straight down to Watchdog Sparky (#10) to trip the safety switch if loops occur!



MAP CONNECTION TO PAGE 22: Plugs into Kernel Supervisor Watchdog (Skill #10)

The purple wire signals Watchdog Sparky (#10). If the 3-ticket hop limit is exceeded, Sparky terminates the loop immediately.

Superpower: Kills endless games of tag so tasks actually get finished!

The 4th Grader Golden Rule:

"3 passes maximum, then you must score the goal!"

PRODUCTION SKILL.MD SPECIFICATION (ENGINEERING CONTRACT)

```
---
name: swarm-hop-limiter
description: Bounded TTL ticket puncher eliminating runaway gos
tier: Level 2 (Multi-Agent Swarms)
ring: Ring 0 (Loop Prevention Engine)
dependencies: [kernel-supervisor-circuit]
tools: [punch_hop_ticket, check_ttl_budget, trip_hop_exha
---
```

1. Purpose & Architectural Boundary

Limits delegation depth with a strict 3-ticket budget, preventing infinite games of tag.

2. Core System Invariants (Non-Negotiable)

- Hard Hop Ceiling: Maximum delegation depth strictly capped at <= 3 hops.
- Monotonic Decrement: Every agent handoff consumes exactly 1 ticket.
- Cycle Detection: Envelope maintains visited node hash list to prevent loo

3. Execution Protocol & Syscalls

1. Stamped message created with budget: ttlBudget = 3.
2. Agent 1 -> Agent 2: Ticket #1 punched (ttl = 2).
3. Agent 2 -> Agent 3: Ticket #2 punched (ttl = 1).
4. If ticket count reaches 0 without task completion: signals Watchdog (#10

4. Failure Modes & Self-Healing

- Zero Tickets Remaining: Trips Watchdog Sparky (#10) to halt loop immediat
- Circular Routing Detected: Drops packet and alerts circuit breaker (#20).

PLAYGROUND MENTAL MODEL (FOR 4TH GRADERS)

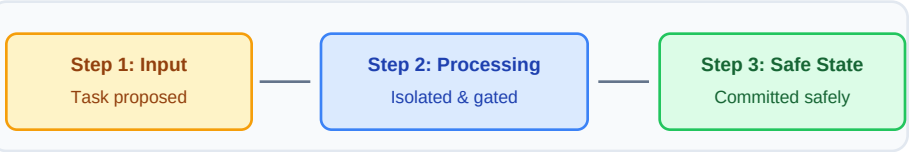
Meet The 3-Column Chalkboard!

WHAT IS THIS SKILL IN ONE SENTENCE?

A big chalkboard at the front of the room divided into 3 neat columns so everyone sees the score without asking!

The Playground Story (How It Works):

Instead of 12 kids writing separate paper notes and throwing them at each other across the classroom, what is much smarter? Draw 3 big columns on the front chalkboard: Column 1 for Money, Column 2 for Rules, Column 3 for Builders! Everyone sees the exact same truth in 1 second! On Page 21's Blueprint, look at the two purple wires dropping from this board: Wire #1 plugs into the Talking Stick (#11) so kids write one at a time, and Wire #2 stamps completed columns into the Wax Diary (#13)!



MAP CONNECTION TO PAGE 22: Plugs into Talking Stick (#11) & Episodic Diary (#13)

Two purple wires drop down: one borrows the Talking Stick (#11) for atomic lane writes, and the other commits snapshots to the Wax Diary (#13).

Superpower: Everyone stays on the same page without sending 100 confusing ema

The 4th Grader Golden Rule:

"Put the score on the board so everyone knows what's happening!"

PRODUCTION SKILL.MD SPECIFICATION (ENGINEERING CONTRACT)

```
---
name: swarm-shared-whiteboard
description: Partitioned O(N) blackboard shared memory with ato
tier: Level 2 (Multi-Agent Swarms)
ring: Ring 0 (Shared State Fabric)
dependencies: [kernel-mutex-lease, kernel-episodic-ledger]
tools: [read_lane_state, write_lane_atomic, seal_whiteboa
---
```

1. Purpose & Architectural Boundary

Replaces O(N^2) point-to-point chatter with an O(N) partitioned blackboard.

2. Core System Invariants (Non-Negotiable)

- Partitioned Lane Ownership: Only Finance writes Finance; only Ops writes
- Atomic Mutex Writes: Updating a lane requires holding Talking Stick (#11)
- Immutable Final State: Finished boards stamped into Wax Diary (#13).

3. Execution Protocol & Syscalls

1. Agents read global board state with zero point-to-point network chatter.
2. When pod has verified result: acquires Mutex lease from Talking Stick (#
3. Commits typed update to designated lane (Finance, Compliance, or Ops).
4. Emits lightweight state_changed event; seals final snapshot to S3 Ledger

4. Failure Modes & Self-Healing

- Write Collision: Backs off and retries via DynamoDB CAS mutex.
- Corrupted Data: Consensus Arbiter (#19) invalidates poisoned entries.

PLAYGROUND MENTAL MODEL (FOR 4TH GRADERS)

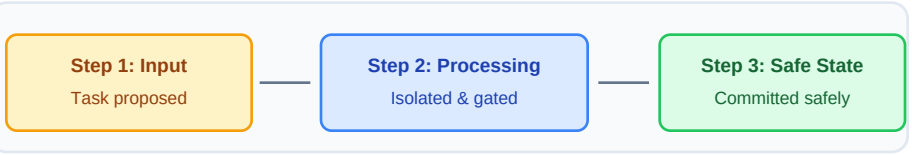
Meet The Super-Vote Ballot Box!

WHAT IS THIS SKILL IN ONE SENTENCE?

A supermajority vote where at least 10 out of 12 teammates must raise their hands before picking the game!

The Playground Story (How It Works):

Imagine your 12-person kickball team is deciding whether to play on Field A or Field B. What if 2 kids are being silly or got tricked? If you needed all 12 kids to agree, 1 grumpy kid freezes recess! The Byzantine Quorum uses the supermajority rule: as long as at least 10 out of 12 kids raise their hands and show verified tickets, the game starts! On Page 21, the purple wire connects this Ballot Box to Detective Triangulator (#14) to verify every ballot with 3 real facts before counting it!



MAP CONNECTION TO PAGE 22: Plugs into Grounding Certifier (Skill #14)

The purple wire verifies ballots with Detective Triangulator (#14). No vote counts unless backed by S3 receipts and vector similarity!

Superpower: Makes fair, unstoppable decisions even if a few robots glitch out

The 4th Grader Golden Rule:

"10 out of 12 agree = game on, no stalling allowed!"

PRODUCTION SKILL.MD SPECIFICATION (ENGINEERING CONTRACT)

```
---
name: swarm-consensus-arbiter
description: Byzantine fault-tolerant supermajority quorum arbi
tier: Level 2 (Multi-Agent Swarms)
ring: Ring 0 (Consensus Engine)
dependencies: [kernel-grounding-certifier]
tools: [collect_ballot, compute_byzantine_quorum, commit_
---
```

1. Purpose & Architectural Boundary

Supermajority consensus (>= 2/3 + 1) ensuring fair decisions even if nodes glitch or lie.

2. Core System Invariants (Non-Negotiable)

- Byzantine Threshold: Requires >= 10 out of 12 votes (>= 2/3 + 1 supermajo
- Cryptographic Proof: Every ballot must carry valid Grounding Certificate
- Liveness Guarantee: Up to 3 faulty or malicious agents cannot stall syste

3. Execution Protocol & Syscalls

1. Swarm coordinator broadcasts high-stakes ballot proposal to all 12 nodes
2. Nodes vote YES/NO accompanied by 3-point fact proof from Detective (#14)
3. Arbiter tabulates verified ballots: requires >= 10 matching votes.
4. Commits binding supermajority verdict to S3 WORM Vault (#3).

4. Failure Modes & Self-Healing

- Quorum Not Reached: Motion rejected; falls back to human escalation.
- Rogue Voting: Faulty agent ballots discarded; agent flagged for quarantin

PLAYGROUND MENTAL MODEL (FOR 4TH GRADERS)

Meet Referee Whistle & Red Card!

WHAT IS THIS SKILL IN ONE SENTENCE?

A safety whistle that benches a runaway robot for 5 minutes so it doesn't bump into all its teammates!

The Playground Story (How It Works):

In soccer, what happens if one player loses their shoes, gets super dizzy, and starts kicking the ball into their own team's net? The referee blows the whistle: *TWEET!* 'Red card! Sit on the bench for 5 minutes and drink some cold water.' On Page 21's Blueprint, look at the two purple wires from this Red Card: Wire #1 alerts Watchdog Sparky (#10) to stop the loop, and Wire #2 forcefully snatches the Talking Stick (#11) away from the dizzy robot!



MAP CONNECTION TO PAGE 22: Plugs into Watchdog Sparky (#10) & Talking Stick (#11)

Two purple wires: one trips Watchdog Sparky (#10) to pause execution, and the other instantly revokes the DynamoDB mutex lease (#11).

Superpower: Protects the rest of the team from one buggy robot causing a disa

The 4th Grader Golden Rule:

"Bench the dizzy player so the rest of the team stays safe!"

PRODUCTION SKILL.MD SPECIFICATION (ENGINEERING CONTRACT)

```
---
name: swarm-circuit-breaker
description: Referee red card bencing faulty agents and revokin
tier: Level 2 (Multi-Agent Swarms)
ring: Ring 0 (Fault Isolation)
dependencies: [kernel-supervisor-circuit, kernel-mutex-lease]
tools: [detect_anomaly, bench_agent, revoke_all_leases]
---
```

1. Purpose & Architectural Boundary

Benches hallucinating or looping agents instantly to prevent cascading team failures.

2. Core System Invariants (Non-Negotiable)

- Instant Quarantine: Benched agents immediately cut off from swarm network
- Mutex Lease Revocation: Forcibly snatches Talking Stick (#11) from benche
- Dynamic Topology Healing: Reroutes liaison bridges around quarantined age

3. Execution Protocol & Syscalls

1. Monitors agent error rates, timeouts, and ungrounded proposal attempts.
2. If error count exceeds threshold (3 strikes): blows red card whistle.
3. Signals Watchdog Sparky (#10) to terminate agent's execution thread.
4. Revokes DynamoDB mutex lease (#11) and quarantines agent on bench for 5

4. Failure Modes & Self-Healing

- Cascade Panic: Quarantines up to 3 nodes independently while maintaining
- Critical Pod Outage: Spawns fresh replacement MicroVM container via Lambd

The Master Blueprint: All 20 Stories & Skills Connect!

Look at the colorful wires! Purple wires connect Playground Swarms (#15-20) to Robot Brains (#8-14). Blue wires connect Robot Brains down to Ground-Floor Cloud Safes (#1-7)!

The 3-Tier Enterprise Agent Stack • Skill Dependency Graph

Unidirectional Architectural Invariants: Level 2 Swarms → Level 1 Kernel OS → Level 0 AWS Cloud Hardware Root

● Level 2: Swarm Coordination (6) ● Level 1: Agent Kernel Harness (7) ● Level 0: Physical AWS Primitives (7)

LEVEL 2 • MULTI-AGENT SYSTEM

Watts-Strogatz Small-World Governance • O(N) Shared Blackboard • Byzantine Quorum

6 Swarm Skills

swarm-topology-builder
Generates small-world lattice (p=0.15) with high clustering and short path lengths.
RING 0 ↓ to kernel-runbook

swarm-liaison-router
Funnels cross-pod queries through dedicated bridge nodes without message flooding.
RING 0 ↓ to propose-decide

swarm-hop-limiter
Enforces strict TTL and delegation budgets (MaxHops ≤ 3) to kill runaway loops.
RING 0 ↓ to supervisor-circuit

swarm-shared-whiteboard
Linear O(N) append-only blackboard state eliminating quadratic O(N²) chat storms.
RING 0 ↓ to episodic & mutex

swarm-consensus-arbiter
Solitary 2/3 + 1 Byzantine fault-tolerant quorum for high-stakes fund/schema actions.
RING 0 ↓ to grounding-cert

swarm-circuit-breaker
Isolates hallucinating nodes: revokes CAS mutex leases and severs graph edges.
RING 0 ↓ to mutex & supervisor

LEVEL 1 • AGENT OS KERNEL

Prefrontal Executive Control Suite • Multi-Store Memory Suite • Virtual Attention Pager

7 Kernel Skills

kernel-propose-decide
Separates untrusted Ring 3 model proposals from Ring 0 execution verdicts.
RING 0 ↓ to bedrock & agentcore

kernel-runbook-engine
Executes structured DAG workflow runbooks with step-barrier verification.
RING 0 ↓ to stepfunctions

kernel-supervisor-circuit
Watchdog monitoring loop divergence, execution timeouts, and token burn budgets.
RING 0 ↓ to lambda & client-api

kernel-mutex-lease
Coordinates atomic CAS leases and background heartbeats with randomized backoff.
RING 0 ↓ to database (DynamoDB)

kernel-context-pager
Virtual memory pager maintaining O(1) active token window with S3 frame eviction.
RING 0 ↓ to s3-vault

kernel-episodic-ledger
Append-only SHA-256 hash-chained block ledger of all perceptions and decisions.
RING 0 ↓ to s3-vault (WORM)

kernel-grounding-certifier
Triangulates assertions across S3 receipts, vector embeddings, and graph triples.
RING 0 ↓ to s3 & database

LEVEL 0 • PHYSICAL AWS CLOUD PRIMITIVES

Hardware Cloud Boundary • Cryptographic WORM Receipts • Isolated MicroVM Sandboxes

7 Cloud Primitives

aws-bedrock
Foundation model inference generating structured JSON proposals. Zero execution IAM.
RING 3 Bedrock Models

aws-agentcore
Ring 0 Cedar guardrails enforcing strict parameter bounds and syscall permissions.
RING 0 Cedar / IAM

aws-s3-vault
Cryptographic WORM storage (Object Lock Compliance) with SHA-256 grounding receipts.
RING 0 S3 Object Lock

aws-database
DynamoDB CAS mutex leasing, OpenSearch Serverless k-NN vectors, and Neptune graph.
RING 0 DynamoDB + Neptune

aws-lambda
Ephemeral stateless compute in memory-capped (512MB/10s) Firecracker microVMs.
RING 1 Firecracker VMs

aws-stepfunctions
Deterministic Hierarchical Finite State Machine with automatic rollback compensation.
RING 0 State Machines

aws-client-api
Perimeter OpenAPI v3 schema validator and Amazon EventBridge inter-cluster event bus.
RING 0 EventBridge + API GW